HAROON **RAZA**

1 x Instrument Engineer | 1 x Trainee Engineer (Embedded Systems) | 4 x Intern (VoIP Engineering, Embedded Systems, Engineering and General)

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EDUCATION

Bachelor of Science, Electrical Engineering September 2021 – July 2025
COMSATS University Islamabad

- Applied core engineering coursework in DLD, Control Systems, and Microprocessor Interfacing to build foundational capabilities in designing, analyzing, and troubleshooting electrical and embedded systems.
- Contributed to the success of PSYWSC'23, an IEEE-sanctioned event, by supporting organization and serving as security professional so event goes smoothly.
- Improved algorithmic problem-solving proficiency by completing the 30 Days of Leetcode Challenge (GDSC), directly enhancing coding readiness for engineering applications.
- Improved team working and leadership skills by leading lab group.

EXPERIENCE

Instrument Engineer March 2026 – Present
VELOSI – Islamabad, Pakistan

- Reviewed the condition of site equipment to ensure asset integrity and verified its physical location.
- Labeled and categorized specific components directly inside images taken on-site.
- Extracted technical data from site photos and files to accurately fill out the Asset Register (AR).
- Performed Engineering Critical Assessment (ECA) checks to evaluate if equipment is safe to keep running.

Trainee Engineer (Embedded Systems) July 2025 – December 2025
CMGL – Islamabad, Pakistan (Remote)

- Enabled robust industrial automation by designing ESP32-based systems.
- Developed a LoRa-based GPS tracking system for real-time location monitoring in remote or low-connectivity environments.
- Implemented a dual-method water level management system between two tanks using both conductive sensing and ultrasonic detection for reliability and redundancy.
- Created IoT based CNC machine which uses Arduino UNO for CNC tasks and ESP32 Cam for controlling CNC remotely, in the meantime streaming video of CNC machine working.
- Ensured reliable long-term operation by using ESP32 NVS and SPIFFS to persistently store calibration data, user settings, and web content across reboots—critical for embedded deployments without user intervention.
- Extended system interoperability by building a universal IR repeater that captures and replays infrared commands across protocols, allowing legacy consumer devices to be controlled programmatically from anywhere via MQTT.
- Improved sensing reliability in water monitoring applications by implementing both conductive probe and ultrasonic methods, enabling flexible deployment across diverse tank types and conditions.

VoIP Engineering Intern February 2025 – April 2025
GOMARKHO – Islamabad, Pakistan

- Enhanced PBX functionality by implementing call routing, interception, and session management in FreeSWITCH, supporting advanced telephony features for business use.
- Accelerated deployment and consistency of VoIP infrastructure by containerizing services with Docker and automating provisioning via CI/CD pipelines on AWS.
- Reduced operational overhead by scripting automated server setup and disk monitoring in Bash, minimizing manual maintenance tasks.

- Strengthened system uptime by maintaining Linux-based VoIP servers with attention to performance and reliability.

Embedded Systems Intern January 2025 – February 2025

Emertxe – Bengaluru, India (Remote)

- Demonstrated real-time embedded control by building a PIC16F877A-based microwave simulator with responsive LCD feedback and keypad input, validated through simulation and firmware testing.
- Improved input reliability by applying debouncing and interrupt handling, ensuring accurate user interaction in time-critical operations.

Engineering Intern June 2024 – August 2024

National Institute of Electronics – Islamabad, Pakistan

- Restored and enhanced dot matrix display control by reverse-engineering and rebuilding its driver library, enabling more precise output and integration.
- Increased the intelligence of a smart parking prototype by contributing to YOLO-based vehicle detection and EasyOCR-based license plate recognition, supporting real-time monitoring.

Operations Manager February 2024 – June 2024

IEEE COMSATS University Islamabad – Islamabad, Pakistan

- Improved team coordination and task completion for IEEE events by adopting G Suite tools, leading to smoother execution and better accountability.

Intern July 2023 – December 2023

Nobel Learning PBC – Gaithersburg, MD, USA (Remote)

- Supported course development by contributing technical content to a "Data Clustering & Visualization" module, helping make complex concepts more accessible to learners.

SKILLS

Technical: Circuit Design, PCB Design, Python, C, C++, Linux, MATLAB, Proteus, Altium, Arduino, IoT, AI, Git, Docker

Analytical: Data Analysis

Communication: Team Communication, Technical Writing

Leadership: Team Leadership, Strategic Planning

Languages: English (Full Professional), Urdu (Native), Punjabi (Native)

PROJECTS

Smart Cart ML Module Project *(Collaborative)*

- Increased accuracy and reduced cost by making barcode scanning the primary item detection method, avoiding the computational overhead and error rates of pure vision-based systems.
- Enabled real-time user engagement by helping design a Raspberry Pi 5 powered touch interface that displays cart contents and personalized recommendations instantly.
- Extended detection capability to new products by implementing a Few-Shot Object Detection (FSOD) model that requires fewer than five training samples per item.
- Improved in-cart support by embedding a RAG-based chatbot that retrieves product information contextually, enhancing the shopping experience.
- Simplified deployment across environments by containerizing all services with Docker, ensuring consistent behavior from development to production.
- Supported sustainability goals by optimizing the system for low power consumption without sacrificing functionality.
- Provided actionable retail insights by building a Django admin dashboard that tracks sales trends and customer behavior.

Smart Home System with ESP32

- Enabled remote control of home appliances by integrating ESP32 with Firebase, allowing users to manage devices from anywhere via mobile app.
- Provided users with full visibility into energy usage by logging and displaying real-time data on temperature, humidity, and power source (solar/grid/generator).
- Reduced generator dependency during outages by implementing intelligent load shifting that prioritizes solar, then grid, then generator—extending fuel and component life.

- Improved maintenance response by adding generator fault alerts directly in the mobile app, reducing downtime.

Line Following Robot

- Achieved precise navigation in constrained paths by tuning a PID controller, demonstrating closed-loop control suitable for real-world robotics applications.

Generator Auto On/Off

- Eliminated manual intervention during power failures by automating generator startup within seconds of grid loss, ensuring uninterrupted power supply.
- Simplified troubleshooting by implementing a self-diagnostic indicator that signals common engine faults, aiding faster repairs.

Intelligent Multi-Source Home Automation System

- Engineered an ESP32-based automation system managing three power sources (WAPDA grid, solar, and backup generator) with automatic failover logic, triggering generator start within seconds of dual-source failure to ensure uninterrupted power supply.
- Implemented CT-based lightweight load monitoring across all three sources to track real-time consumption and enable informed switching decisions.
- Integrated DHT22 environmental sensors for temperature/humidity monitoring which could be used with conditional appliance control (e.g., fan activation on thermal thresholds).
- Developed MIT App Inventor mobile interface connected via Firebase Realtime Database, providing live telemetry (temperature/humidity graphs, generator/pump status) and remote control of 4+ appliances with sub-second synchronization.

CERTIFICATIONS

- Introduction to R – ***GreatLearning***
- Programming Essentials in C++ – ***CISCO***
- Python Essentials – I – ***CISCO***
- Python Beyond the Basics – ***10 Pearls***
- 30 Days of Leetcode Challenge (Oct 2023) – ***GDSC COMSATS***
- Introduction to Programming with MATLAB – ***Coursera***
- MATLAB Programming for Engineers and Scientists Specialization – ***Coursera***
- AI Agents Course – ***Hugging Face***
- Applied Data Science Lab – ***WorldQuant University***

References available upon request | Last Updated: June 12, 2026